

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	15 February 2026
Team ID	LTVIP2026TMIDS88090
Project Name	AUTOSAGE APP USING GEMINI FLASH
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

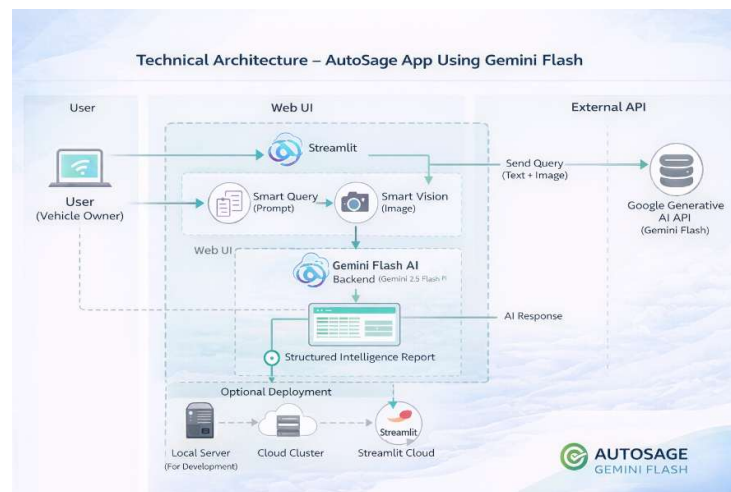


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based interface where users submit vehicle queries (text/image) and view structured intelligence reports.	Streamlit (Python), HTML, CSS
2.	Application Logic-1	Handles user inputs, prompt construction, validation, and report formatting logic	Python
3.	Application Logic-2	Multimodal AI processing for vehicle analysis (text + image understanding)	Google Gemini 2.5 Flash (Generative AI API)
4.	Application Logic-3	Vehicle intelligence structuring, powertrain determination rules, Indian market contextualization	Custom Prompt Engineering + Python Processing
5.	Database	(Optional) Store user queries, analysis history, logs.	SQLite / PostgreSQL (Optional)
6.	Cloud Database	Deployment and hosting environment	Streamlit Cloud / Local Server.
7.	File Storage	Temporary storage of uploaded vehicle images for processing	Local Filesystem (Streamlit Upload Handler)
8.	External API-1	AI model API for generating structured vehicle intelligence reports	Google Generative AI API (Gemini Flash)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Web application framework and supporting libraries used for building UI and backend logic	Streamlit, Python, Pillow, dotenv
2.	Security Implementations	Secure API key management, environment variable protection, HTTPS-based API communication, input validation	Environment Variables (.env), HTTPS (TLS), Google API Security, Basic Input Validation
3.	Scalable Architecture	Modular architecture separating UI, Prompt Engine, and AI Processing Layer. Can scale horizontally when deployed on cloud platforms	Streamlit Deployment + Google Gemini Cloud API (Cloud-based scalable inference)
4.	Availability	Modular architecture separating UI, Prompt Engine, and AI Processing Layer. Can scale horizontally when deployed on cloud platforms	Streamlit Cloud / Cloud Hosting + Google Cloud Infrastructure
5.	Performance	Optimized prompt engineering to reduce token usage; lightweight UI; cloud-based AI processing; controlled temperature & max token configuration	Gemini 2.5 Flash (Low Latency Model), Python Optimization, Configured Token Limits